MATH 11 — ELEMENTARY STATISTICS

§1.3

Levels of Measurement

Which operations on the data actually mean something

Today's Goal

After completing this section, you should be able to:

- Identify the four levels of measurement: nominal, ordinal, interval, and ratio
- Distinguish between each level of measurement
- Determine the appropriate level of measurement for a given dataset

What You Will Be Graded On

LT.2 Levels of Measurement

I understand the four levels of measurements. I can use this understanding to identify what level of measurement is appropriate for a given data set and can also give examples of data sets for each level of measurement.

Which of These Can You Average?

Eye color. Letter grades. Calendar years. Height.

Height, certainly. Calendar years, also yes. Letter grades, doubtful. Eye color, never.

The data type does not settle this. Something finer does.

Names With No Order

Definition 1.3.1

Data is at the **nominal level** of measurement if it consists of names, labels, or categories, and the data cannot be arranged in any meaningful order.

Eye color, gender, political party affiliation, and zip codes.

All you can ask of two values is whether they are the same or different.

Order, But No Measurable Gaps

Definition 1.3.2

Data is at the **ordinal level** of measurement if it can be arranged in a meaningful order, but differences (obtained by subtraction) between data values either cannot be determined or are meaningless.

Rankings, letter grades, and survey scales from strongly agree to strongly disagree.

Gaps That Mean Something

Definition 1.3.3

Data is at the **interval level** of measurement if it can be arranged in a meaningful order, differences between data values are meaningful, but there is no natural zero point (a value of zero does not indicate the absence of the quantity).

Temperature in Fahrenheit or Celsius, calendar years, and IQ scores.

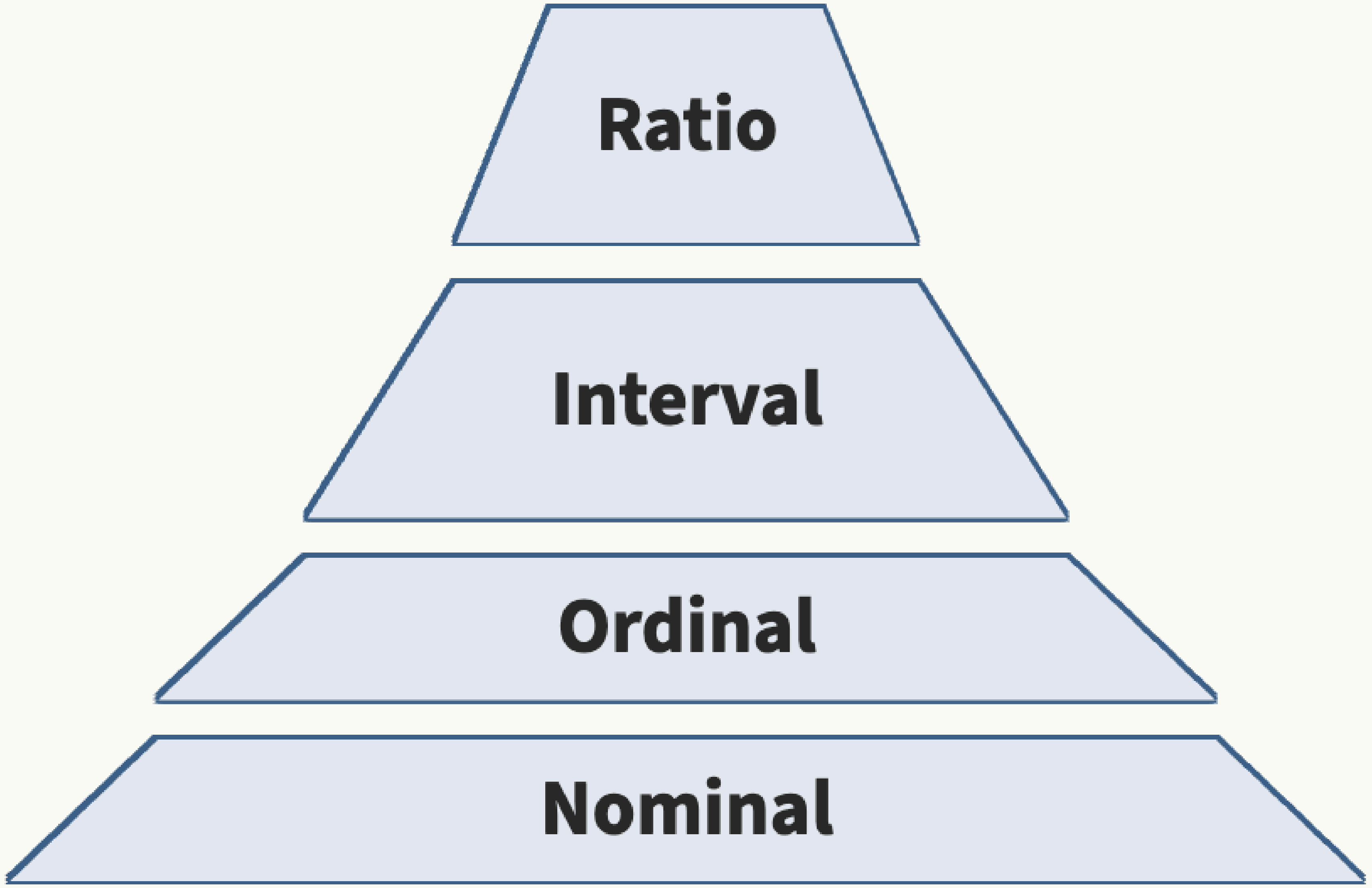
A Zero That Means None

Definition 1.3.4

Data is at the **ratio level** of measurement if it can be arranged in a meaningful order, differences between data values are meaningful, and there is a natural zero point (a value of zero indicates the absence of the quantity).

Height, weight, age, income, and distance.

Each Level Adds One Thing



Adds a true zero and meaningful ratios ($\times$ or $\div$)

Adds meaningful differences (subtraction)

Adds meaningful order ($<$ or $>$)

Categories only ($=$ or $\neq$)

The Four Levels Side by Side

Level	Order?	Differences?	True Zero?	Ratios?	Examples
Nominal	No	No	No	No	Eye color, zip codes, SSN
Ordinal	Yes	No	No	No	Rankings, letter grades
Interval	Yes	Yes	No	No	Temperature (°F), years
Ratio	Yes	Yes	Yes	Yes	Height, weight, age, income

Is 90°F Twice as Warm as 45°F?

The numbers say one is twice the other. Does the weather?

It does not. Interval and ratio are the two levels students mix up, and this is where they come apart.

Two tests settle it, and they always agree.

Two Tests That Always Agree

Ratio Test. Does it make sense to say one value is “twice as much” as another?

- If yes, the data is at the **ratio** level.
- If no, the data is at the **interval** level.

True Zero Test. Does a value of zero indicate the complete absence of the quantity?

- If yes, the data is at the **ratio** level.
- If no, the data is at the **interval** level.

Putting Fahrenheit Through the First Test

Yesterday in Fresno the low was 45°F and the high was 90°F.

Ratio Test. Was the afternoon twice as hot as the morning?

The same low and high in Celsius: 7.2°C and 32.2°C — about 4.5 to 1, not 2 to 1.

Two scales, two different ratios for the same two moments. A real doubling could not depend on which scale you wrote it in. Something is wrong with the scale itself — and the second test says what.

And Through the Second Test

True Zero Test. Does 0°F mean there is no temperature, no heat at all?

No. 0°F is just another point on the scale. It is a cold day, not the absence of heat.

Both tests agree: temperature in Fahrenheit is at the **interval level**.

Kelvin is the exception. 0 K is absolute zero, the complete absence of thermal energy, so Kelvin is at the **ratio level**.

Naming the Level, Part One

For each of the following, determine which of the four levels of measurement best describes the data.

1. A campus survey records the home county of each new student. **Nominal.** Counties are categories with no meaningful order.
2. A restaurant asks each diner to rate the service as Poor, Fair, Good, or Excellent. **Ordinal.** The ratings can be ordered, but the gap from Poor to Fair need not match the gap from Good to Excellent.

Naming the Level, Part Two

3. A school psychologist records the IQ score of each student tested. **Interval.** Scores can be ordered and differences are meaningful, but an IQ of 0 does not mean no intelligence, so 140 is not twice 70.
4. A delivery company records the weight of each package, in pounds. **Ratio.** A weight of 0 pounds means no weight at all, so a 10 pound package really is twice a 5 pound package.

Numbers That Are Not Numbers

One player wears jersey number 23. A teammate wears 46. Is 46 twice 23 in any useful sense?

No. The numbers name players. That makes them nominal, however they are written.

Now You Try: Name the Level

You Try 1.3.1. For each of the following, determine which of the four levels of measurement best describes the data.

1. A hospital records the resting heart rate of every patient admitted. **Ratio.**
2. A bank records the Social Security number of each new account holder. **Nominal** — digits, but only a label.
3. A student consults published rankings of medical schools. **Ordinal.**
4. The number of hours each employee worked last week. **Ratio.**

Averaging a Survey Scale

Survey scales from 1 to 5 get added and averaged all the time. You will see it in reports, in product reviews, and in course evaluations.

At the ordinal level, that arithmetic is not strictly proper. The gap from 1 to 2 need not equal the gap from 4 to 5.

The practice is common. Knowing why it is questionable is the point of this section.